

RLBA-UAV: A Robust and Lightweight Blockchain-Based Authentication and Key Agreement Scheme for PUF-Enabled UAVs

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Abstract—Unmanned aerial vehicles (UAVs) integrated with the internet of things (IoT) guarantee useful advantages such as facilitating ground communications in regions where the availability of connectivity is restricted owing to physical obstacles. However, the data transmitted by sensors and IoT embedded in UAVs are facing new security issues and privacy challenges with the known security attacks over time. To address these security attacks and threats and meet lightweight UAV communication requirements, a secure and lightweight authentication and key agreement (AKA) scheme is essential. Recently, researchers have designed a lightweight blockchain-enabled AKA scheme with privacy-preserving for UAVs to provide useful and reliable services. However, we prove that the existing scheme is fragile to various security attacks and does not ensure mutual authentication. Thus, we propose a robust and lightweight blockchain-based AKA scheme for PUF-enabled UAVs, called RLBA-UAV to enhance the security problems of the existing scheme. We demonstrate the security of RLBA-UAV by using informal/formal security analyses such as the ROR oracle model and AVISPA simulation. Moreover, we demonstrate the performance comparison analysis between RLBA-UAV and related schemes for UAVs. We demonstrate an implementation of a network simulator (NS) 3 compliant with IEEE 802.11p standards to show its validation and feasibility that RLBA-UAV is appropriate for practical UAVs. Thus, RLBA-UAV offers enhanced security and operational efficiency compared to related schemes and can be applied to practical blockchain-based AKA systems for UAVs.

Index Terms—Unmanned aerial vehicles (UAVs), blockchain, physical unclonable functions (PUF), authentication.

I. INTRODUCTION

WITH the development of 5G communications and mobile devices, unmanned aerial vehicles (UAVs) with

the internet of things (IoT) provide numerous advantages, including facilitating ground communications in regions with restricted connectivity caused by physical obstacles [1], [2], [3]. UAVs offer back-end services such as low-latency, fast, and intelligent network properties in smart cities. In addition, UAVs perceive and collect intelligently various environmental information through the built IoT sensors, including the location from a GPS, air pressure from a barometer, orientation from a gyroscope, and videos and photographs from cameras. Based on the environmental information, UAVs provide useful functionalities and services such as monitoring, transportation, rescue, and surveillance in smart cities, and have substantially enhanced the quality of citizen's life in recent years [4]. For instance, if there is an accident or a natural calamity, UAVs can gather data using distance imaging and thermal sensors. This enables them to precisely determine the locations of missing individuals and injured victims even in extreme conditions or challenging landscapes. Subsequently, the remote paramedics and firefighters can access the data acquired by UAVs in real-time to closely monitor the situation and respond promptly. However, despite the numerous advantages offered by UAVs, there are significant issues and challenges [5]. If the data collected from UAVs is damaged or leaked, a malicious adversary can attempt various security threats and even exploit the information for illegal purposes. For instance, a malicious adversary can focus on specific local areas of importance to capture images, secretly film, or deploy UAVs to carry out to drop a bomb. Furthermore, an adversary can obtain the secret credentials from a physically captured UAV and then impersonate a legitimate UAV by using the stolen data. To resolve these problems, a physical unclonable function (PUF) [6] has emerged as a promising solution to guarantee high security and efficiency suitable for UAV's requirements since UAVs generally have low-computing and limited resources.

With the growing demand for services and applications in smart cities, another significant concern of UAVs is a lightweight attribute. The rapid expansion of UAVs poses several issues and challenges in handling large amounts of data exchanged between UAVs, ground station servers, and users [7]. UAVs are incapable of executing high-complexity computations because they are constrained in terms of energy consumption and computing power and also the operational system for UAVs requires that the computation should be completed within a limited time. Otherwise, the performance

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efficiency will be greatly affected. Thus, it is imperative to guarantee the security of this data while simultaneously providing effective performance costs. Thus, many researchers have proposed numerous lightweight authentication and key agreement (AKA) schemes for UAVs to improve these problems [8], [9]. While the majority of these have been demonstrated to be valid to a certain degree, a ground station server or a trusted authority (TA) remains computation/communication overheads are still unresolved issues currently. Besides, the database of trusted servers cannot solve issues such as single point of failure (SPOF) and bottlenecks since it is a centralized system. To address these issues, a promising solution is the use of blockchain technology. The blockchain is a state-of-the-art network technology that ensures decentralization and data integrity to the seamless exchange of information with multiple distributed nodes. The blockchain guarantees a decentralized and secure type of distributed ledger technology and offers significant advantages for scenarios related to UAVs. Although these blockchain-enabled UAV systems have received significant attention, these schemes are still fragile to potential security threats.

Recently, Wang et al. [10] proposed a lightweight blockchain-enhanced AKA scheme with privacy-preserving for UAVs to provide effective and reliable services. Wang et al. asserted that their AKA scheme guarantees essential security prerequisites while preventing severe security attacks. However, we demonstrated that their AKA scheme cannot prevent severe security attacks such as “impersonation and session key disclosure” attacks and lacks “mutual authentication”. Therefore, we propose a “robust and lightweight blockchain-based AKA scheme for PUF-enabled UAVs, called RLBA-UAV” to resolve the security weaknesses of [10].

A. Contributions

We introduce the main contribution of RLBA-UAV.

- We propose a “robust and lightweight blockchain-based AKA scheme for PUF-enabled UAVs” to improve the security issues of [10].
- We perform “Automated Verification of Internet Security Protocols and Applications (AVISPA) [11]” to evaluate the robustness against lethal security attacks such as “man-in-the-middle (MITM)” and “replay” attacks. Furthermore, we perform “Real-or-Random (ROR) oracle” model [12] to prove a session key security of RLBA-UAV.
- We provide the implementation for the performance evaluation employing the “network simulator (NS) 3” [13] in compliance with “IEEE 802.11p standard” across a range of flying ad-hoc network (FANET) scenarios.
- We demonstrate the effectiveness by performing test-bed experiments on “multiprecision integer and rational arithmetic cryptographic library (MIRACL)” [14]-enabled Raspberry PI 4. Based on the experiments, we prove that RLBA-UAV guarantees efficient performance as compared to previous schemes in the literature.

B. Paper Outline

The rest of the paper is organized as follows. Section II and III introduce the related works for UAV environments

and preliminaries. In Section IV and V, we prove the security weaknesses of Wang et al. scheme [10] and then proposes a robust and lightweight blockchain-based authentication and key agreement scheme for PUF-enabled UAVs, called RLBA-UAV to enhance to resolve the security flaws of Wang et al. scheme [10]. Section VI analyzes the security and robustness of RLBA-UAV by performing informal and formal security analyses. Section VII presents the testbed experiments for cryptographic primitives using MIRACL-based Raspberry PI4 and Desktop. In Section VIII, we demonstrate the performance comparative analysis of the proposed scheme with related schemes for UAVs. Section IX provides the implementation for network communication performance using NS 3 in compliance with IEEE 802.11p. Finally, Section X summarizes the conclusions of the paper.

II. RELATED WORKS

Secure and dependable communication stands out as a crucial prerequisite for UAVs to ensure secure data exchange and collaboration. Therefore, an imperative aspect of UAV is the implementation of a resilient authentication and key establishment mechanism, which guarantees secure and effective data interchange between the various components.

The development of a lightweight AKA scheme for UAV systems has garnered substantial attention due to the efficiency concerns of the PKC-based scheme, particularly about sensor devices and IoT with constrained resources. The lightweight scheme is commonly categorized into two types: “two-factor or three-factor” schemes. These schemes employ lightweight cryptographic operations such as “XOR function”, “hash function”, “symmetric key encryption/description” to provide optimal efficiency.

Over the last few years, numerous researchers have designed robust and lightweight AKA schemes for UAVs [15], [16], [17]. Wazid et al. [18] proposed a three-factor-based robust AKA scheme for UAVs. They introduced the necessary security attributes along with the several security challenges for UAVs. Regrettably, Alladi et al. [19] identified that Wazid et al. scheme [18] fails to provide perfect backward secrecy and independent aliasing and also cannot prevent impersonation attack. Alladi et al. [19] presented a two-stage lightweight AKA scheme for software-defined networking (SDN)-enabled UAVs. However, Beebak et al. [20] identified that Alladi et al. scheme [19] cannot resist offline password guessing, session key disclosure, replay, and forgery attacks and also does not guarantee perfect forward secrecy and confidentiality. Srinivas et al. [21] introduce a lightweight three-factor AKA scheme for UAV environments based on temporal credentials. However, Ali et al. [22] pointed out that Srinivas et al. scheme [21] cannot prevent stolen verifier and impersonation attacks and also does not guarantee untraceability. Ali et al. [22] devised a robust AKA for UAV-based smart cities to enhance the security deficiencies identified in Srinivas et al.’s scheme [21]. Unfortunately, Ali et al.’s scheme [22] is still susceptible to server spoofing, forgery, and session key disclosure attacks. Nikooghadam et al. [23] proposed a provably secure and lightweight AKA scheme for UAV to guarantee the smart-life of citizens in smart cities. However, Nikooghadam et al. scheme [23] is still vulnerable to replay,

impersonation, and insider attacks and also does not guarantee mutual authentication. These AKA schemes provide various advantages such as computing power capability and flexibility for UAV communication on centralized cloud servers, but they raise various information security and privacy issues during transmission. Moreover, the cloud-based AKA schemes for UAVs [18], [19], [20], [21], [22], [23] are essentially a centralized system so they do not resolve issues such as single point of failure (SPOF) and bottlenecks. Hence, blockchain technology which possesses decentralized characteristics is considered as a prospective solution to address the problems associated with centralized systems.

Recently, many blockchain-based schemes for UAVs have been designed [24], [25], [26] to address the problems associated with centralized cloud server. Mehta et al. [27] have presented an extensive overview of the security and privacy issues to provide valuable services to users within blockchain-assisted UAV environments. Furthermore, a review of the drawbacks associated with existing UAV systems was conducted, alongside the presentation of viable resolutions to be implemented in the next generation of UAVs. Yazdinejad et al. [28] designed a blockchain-enabled UAV system architecture. This architecture aims to provide users with a communication mechanism that is both transparent and efficient, with a strong emphasis on security. However, these existing blockchain-based schemes for UAVs [27], [28] have garnered considerable interest, these schemes have not been proposed with more detailed blockchain-based AKA protocol for UAVs. Thus, Wang et al. [10] designed a detailed dependable, and lightweight blockchain-enhanced AKA protocol for UAVs. They claimed that their scheme is resilient to potential security threats and also does provide necessary security requirements. However, we identified that Wang et al. scheme [10] is fragile to ground station “server impersonation” and “session key disclosure” attacks. Thus, we design a “robust and lightweight blockchain-based AKA scheme for PUF-enabled UAVs” to resolve the security weaknesses of Wang et al. scheme [10].

RLBA-UAV utilizes smart contracts to ensure efficient access control, maintain data integrity, and achieve decentralization. In RLBA-UAV, each block contains only metadata to optimize block construction and reduce distributed storage waste. RLBA-UAV enables the generation of secret keys using PUF without storing sensitive data in memory. As a result, even if an adversary captures the UAV or mobile device, the adversary cannot be impersonated as a legitimate entity since secret credentials cannot be extracted under the proposed scheme. In RLBA-UAV, service providers can guarantee the integrity and privacy of sensitive data, and mobile users benefit from more efficient and reliable UAV services.

III. PRELIMINARIES

A. Threat Model

We present the “Dolev-Yao (DY) threat” model [29]. The adversary capabilities in security protocol are as follows:

- An adversary ($\mathcal{A}$) can “block, intercept, delete, eavesdrop, resend, and insert” the exchanged data over an open channel and then attempt various security attacks.

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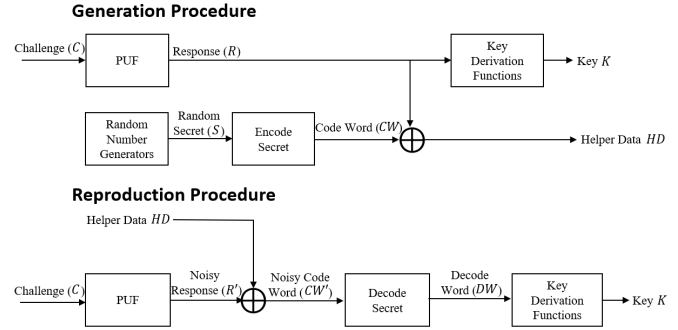


Fig. 1. PUF key generator mechanism.

- $\mathcal{A}$ can steal a mobile device (MD) of the legal user and subsequently retrieve the information contained within MD using “differential power analysis” [30].
- When a UAV is captured by a malicious adversary, the adversary may endeavor to calculate a session key of another uncompromised UAV or a legitimate user using power-analysis attacks [31].

B. Physical Unclonable Function

PUF is widely adopted as a actual solution for safeguarding the security of IoT and UAV devices with constrained computing and memory capabilities from severe security threats [32]. PUF is a “technique for generating an output based on input like a fingerprint obtained from the physical microstructure of smart devices”. PUF does not maintain a secret key in the memory and even if the adversary tries to generate an identical PUF, they have significant difficulties. The reason is due to the complex nanoscale fluctuations that occur during the manufacturing procedure of the integrated circuit (IC) chip. The ideal PUF ensures essential attributes including unpredictability, reliability, and uniqueness, crucial for fortifying the security of smart devices. PUF is a robust defense mechanism, especially adept at fortifying Unmanned Aerial Vehicles (UAVs) integrated into blockchain-operated systems, mitigating diverse physical security threats such as side-channel vulnerabilities, tampering endeavors, and cloning exploits. Moreover, PUF facilitates the validation of the legitimacy of entities before establishing a session key, as indicated in prior scholarly investigations [33]. The functionalities of the PUF are delineated as follows:

- PUF is rather simple to assess and implement.
- PUF relies on the physical microstructure of the systems.
- Any attempt to manipulate smart devices equipped with PUF will update PUF’s behavior and thereby its destruction [34].

As illustrated in Fig. 1, a PUF-based key generator process involves multiple functions such as “encoding, key derivation, and decoding” to produce powerful extractors for a private secret key. The amalgamation of these functions presents an optimal solution for ensuring robust authentication of lightweight smart devices within blockchain-integrated UAV operational ecosystems.

C. Network Model

We present the blockchain-based network model for PUF-enabled UAVs in Fig. 2. The proposed network model

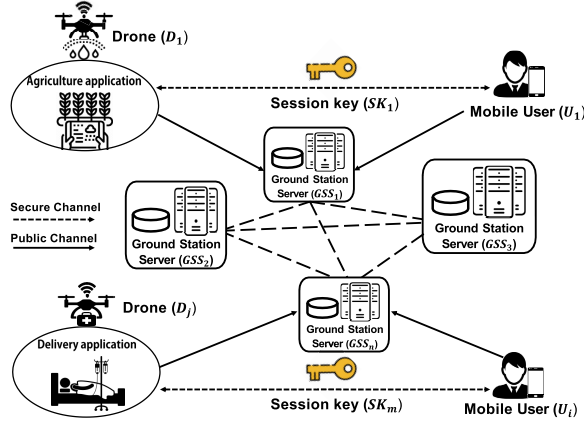


Fig. 2. Blockchain-based network model for UAVs.

has consisted three entities of the “mobile user (U_i), drones (D_j), and connected ground station server (GSS_n)”.

In the network model, the participated nodes in the blockchain are composed of GSS_n . The GSS manages the deployed UAVs within their respective flying zones. GSS collects real-time data from the UAVs, records it as transactions using smart contracts, and then shares it with the blockchain network. Thus, RLBA-UAV allows only authenticated participants to outsource UAV data, and each operation for outsourcing data is integrated into the blockchain as a transaction. Moreover, RLBA-UAV draws valuable conclusions from the stored, processed, and analyzed UAV data to it provides the mobile users. For example, it will predict the chances of disaster, road conditions, etc.

- U_i : This entity carries a mobile device and requests registration from GSS_n and then receives the secret credentials related to authentication in the registration phase. Then, U_i performs mutual authentication with each entity and establishes a common session key SK between U_i and D_j for future secure communications.
- D_j : The secret credentials are provided to D_j through the assistance of the GSS_n and registered in blockchain network prior to its deployment in the designated airspace. The deployed D_j within the designated airspace is remotely managed and controlled and acquires data from various applications via sensors and shares the gathered data.
- GSS_n : Each GSS_n is connected to each other to form a blockchain network, and the GSS_n continuously supervises each other. If any GSS_n is considered vulnerable, other nodes will pause and scrutinize it. In addition, GSS_n serves as a bridge between the D_j and U_i , and D_j and U_i mutually authenticated over an open channel with the help of the GSS_n . The sensitive data for identity authentication is stored within the the smart contract SC and itself rather than the GSS_n 's database. Moreover, an adversary ($\mathcal{A}$) cannot obtain these secret credentials including the previous deployer because the modifier of sensitive data is set as private.

IV. SECURITY WEAKNESSES OF WANG ET AL.'S SCHEME

This section proves the security weaknesses based on the threat model inherent in scheme designed by Wang et al [10]. In Table I, We illustrate the symbols utilized in the article.

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TABLE I
SYMBOLS

Symbol	Meaning
U_i, D_j, GSS_n	i^{th} user, j^{th} drone, n^{th} ground station server
ID_u, ID_d, ID_{gss}	Identity of U_i, D_j, GSS_n
PID_u, PID_d, PID_{gss}	Pseudo-identity of U_i, D_j, GSS_n
PW_u	Password of U_i
SC	Smart contract
R_i	Random numbers
RN_i	Random nonces
$Auth$	Authentication messages
X_i	Secret credentials between MU_i and GSS_n
Z_j	Secret credentials between D_j and GSS_n
C_u^x, R_u^x	Challenge/response of U_i
C_d^x, R_d^x	Challenge/response of D_j
ΔT_i	Acceptable time delay
TS_i	Timestamp
MK_u, MK_d, MK_{gss}	Secret key of U_i, D_j, GSS_n
SK	A session key among U_i and D_j
$E_K(\cdot)/D_K(\cdot)$	Symmetric key encryption/decryption
$h(\cdot)$	Hash function
$\oplus$	XOR function
$ $	Concatenation

A. Session Key Disclosure Attack

According to informal security analysis (i.e., stolen mobile device attack) of Wang et al.'s scheme, $\mathcal{A}$ can extract the secret credentials $\{ID_d, RPW_u, R_2, PID_d, EC_d, PID_{gss}\}$ stored in MD . Moreover, $\mathcal{A}$ can eavesdrop, delete, insert, block, and resend the transmitted data $\{M_1, M_2, M_3, TS_1\}$, $\{M_4, M_5, M_6, M_7, TS_2\}$, and $\{M_8, M_9, TS_3\}$ over a public channel based on the DY threat model. Thus, $\mathcal{A}$ obtain a session key of the legitimate D_j .

In this attack, $\mathcal{A}$ obtains $PID_u = M_1 \oplus h(PID_{gss}||TS_1)$, $R_4 = M_5 \oplus h(EC_d||PID_u||PID_d||PID_{gss}||TS_2)$, $EC_u^{new} = M_6 \oplus h(EC_d||PID_u||PID_d||PID_{gss}||TS_2)$. Then, $\mathcal{A}$ computes $SK_{D-U} = h(PID_u||PID_d||PID_{gss}||R_4||EC_u^{new}||TS_3)$. Consequently, Wang et al.'s scheme cannot prevent the session key disclosure attack because $\mathcal{A}$ calculate SK_{D-U} successfully.

B. GSS Impersonation Attack

According to the adversary model III-A, $\mathcal{A}$ can extract the secret credentials $\{ID_d, RPW_u, R_2, PID_d, EC_d, PID_{gss}\}$ stored in MD to attempt various security threats. Moreover, $\mathcal{A}$ can obtain the exchanged data over a public channel based on the DY threat model. Thus, $\mathcal{A}$ tries to impersonate the GSS_n as follows.

- **GIA-1:** U_i computes $M_1 = h(PID_{gss}||TS_1) \oplus PID_u$, $M_2 = h(PID_u||PID_{gss}||EC_u||TS_1) \oplus PID_d$, and $M_3 = h(PID_u||PID_{gss}||PID_d||EC_u||TS_1)$, where TS_1 is current timestamp. Then, U_i constructs a transaction with message $\{M_1, M_2, M_3, TS_1\}$ to GSS_n . After that, U_i transmits its to the GSS_n over a public channel.
- **GIA-2:** $\mathcal{A}$ eavesdrops the message $\{M_1, M_2, M_3, TS_1\}$ transmitted from U_i and calculates $PID_u = M_1 \oplus h(PID_{gss}||TS_1)$. Then, $\mathcal{A}$ generates a random number R_A and computes $EC_A^{NEW} = h(EC_u||R_A)$, $M_4 = h(EC_d||PID_d||PID_{gss}||TS_2) \oplus PID_u$, $M_{5A} = h(EC_d||PID_u||PID_d||PID_{gss}||TS_2) \oplus R_A$, $M_{6A} = h(EC_d||PID_u||PID_d||PID_{gss}||TS_2) \oplus EC_A^{NEW}$, and $M_{7A} = h(EC_d||PID_u||PID_d||PID_{gss}||$

$TS_2||EC_A^{NEW}$). Then, $\mathcal{A}$ transmits the message $\{M_4, M_{5A}, M_{6A}, M_{7A}, TS_2\}$ to the D_j .

- **GIA-3:** After receiving the message, D_j verifies whether of $|TS_2^* - TS_2| \leq \Delta T_i$. If TS_2 is valid, D_j calculates $PID_u = h(EC_d||PID_d||PID_{gss}||TS_2) \oplus M_4$, $R_A = h(EC_d||PID_u||PID_d||PID_{gss}||TS_2) \oplus M_{5A}$, $EC_A^{new} = M_{6A} \oplus h(EC_d||PID_u||PID_d||PID_{gss}||TS_2)$ and $M_{7A}^* = h(EC_d||PID_u||PID_d||PID_{gss}||TS_2||EC_A^{new})$. After that, D_j checks $M_{7A}^* \stackrel{?}{=} M_{7A}$. If it is not equal, the current session is terminated, otherwise D_j establishes a session key $SK_{ADU} = h(PID_u||PID_d||PID_{gss}||R_A||EC_A^{NEW}||TS_3)$ and computes $M_{8A} = h(PID_u||PID_d||PID_{gss}||TS_3) \oplus R_A$, and $M_9 = (SK_{ADU} \oplus TS_3)$. Finally, D_j sends the message $\{M_{8A}, M_{9A}, TS_3\}$ to U_i over a public channel.
- **GIA-4:** After getting the message, U_i checks the timestamp of $|TS_3^* - TS_3| \leq \Delta T_i$. If it is equal, U_i calculates $R_A = M_{8A} \oplus h(PID_u||PID_d||PID_{gss}||TS_3)$, $SK_{AUD} = h(PID_u||PID_d||PID_{gss}||R_A||EC_A^{new}||TS_3)$. If $(SK_{AUD} \oplus TS_3 = M_{9A})$, $SK_{AUD}(SK_{ADU})$ is considered as legitimate.

Consequently, Wang et al.'s scheme is not resilient to impersonation attacks since $\mathcal{A}$ successfully can impersonate as the GSS_n .

C. Mutual Authentication

Wang et al. claimed their scheme offers “secure mutual authentication”. However, according to Section IV-A and IV-B, $\mathcal{A}$ successfully can generate the authentication request message $M_3 = h(PID_u||PID_{gss}||PID_d||EC_u||TS_1)$, $M_7 = h(EC_d||PID_u||PID_d||PID_{gss}||TS_2||EC_u^{NEW})$, and response message $M_9 = (SK_{D-U} \oplus TS_3)$ for mutual authentication. Consequently, Wang et al.'s scheme does not guarantee “secure mutual authentication” between U_i , GSS_n , and D_j .

V. PROPOSED SCHEME

We design a “robust and lightweight blockchain-based AKA scheme for PUF-enabled UAVs (RLBA-UAV)” to enhance the security flaws of [10].

A. Initial Setup Phase

RLBA-UAV adopts the initial setup phase presented in Wang et al.'s scheme.

B. Drone Registration Phase

D_j should be registered in the blockchain network before formal deployment. This phase process is invoked by the *DroneRegister* function pre-defined in SC . When the *DroneRegister* function is called, GSS_n will support SC to execute the next steps.

- **DRP-1:** GSS_n first calls *DroneRegister* function in SC to help D_j register in blockchain network.

Mobile User (MU_i)	Ground Station Server (GSS_n/SC)
Selects a unique ID_u and PW_u Generates a random number R_2 Computes $RPW_u = h(PW_u R_2)$ $\{ID_u, RPW_u\}$ (via secure channel)	Computes Selects a random number R_3 Selects a PUF set of (C_u^x, R_u^x) Computes $PID_u = h(ID_u R_3)$ $X_i = h(ID_u R_3 R_u^x)$ $Q_i = X_i \oplus h(PID_u RPW_u R_u^x) \oplus RPW_u$ $W_i = h(PID_u RPW_u R_u^x X_i)$ Stores $\{PID_u, X_i, (C_u^x, R_u^x)\}$ $\{PID_u, PID_{gss}, PID_d^{all}, (C_u^x, R_u^x), Q_i, W_i\}$ (via secure channel)
Computes $E_i = MK_u \oplus (PID_u, PID_{gss}, PID_d^{all}, R_2, (C_u^x, R_u^x), Q_i, W_i)$ Stores $\{E_i\}$ in the memory	

Fig. 3. Mobile user registration process of RLBA-UAV.

- **DRP-2:** SC selects a random number R_1 , the identity ID_d , and a PUF set of (C_d^x, R_d^x) . Then, SC computes the pseudo-identity $PID_d = h(ID_d||R_1)$, $PID_{gss} = h(ID_{gss}||R_1)$, and $Z_j = h(ID_d||R_1||R_d^x)$.
- **DRP-3:** SC stores $\{PID_d, Z_j, PID_{gss}, (C_d^x, R_d^x)\}$ securely and sends them to the D_j over a secure channel.
- **DRP-4:** D_j receives $\{PID_d, (C_d^x, R_d^x), Z_j, PID_{gss}\}$ and calculates $N_j = (PID_d, (C_d^x, R_d^x), Z_j, PID_{gss}) \oplus MK_d$. Finally, D_j securely stores $\{N_j\}$ in the memory.

C. User Registration Phase

U_i duly registers in the GSS_n by calling the *UserRegister* function pre-defined in SC . This function issues specific secret credentials for authentication to the U_i over a secure channel. This phase is presented as shown in Fig. 3 and the system participants conduct the following steps.

- **URP-1:** U_i first selects the identity ID_u , password PW_u , and random number R_2 . Then, U_i computes $RPW_u = h(PW_u||R_2)$ and calls *UserRegister* function with the credentials $\{ID_u, RPW_u\}$.
- **URP-2:** After getting the message, SC selects a random number R_3 and a PUF set of (C_u^x, R_u^x) and calculates the pseudo-identity $PID_u = h(ID_u||R_3)$, $X_i = h(ID_u||R_3||R_u^x)$, $Q_i = X_i \oplus h(PID_u||RPW_u||R_u^x) \oplus RPW_u$ and $W_i = h(PID_u||RPW_u||R_u^x||X_i)$.
- **URP-3:** SC stores $\{PID_u, X_i, (C_u^x, R_u^x)\}$ securely and transmits $\{PID_u, PID_{gss}, PID_d^{all}, (C_u^x, R_u^x), Q_i, W_i\}$ to the U_i over a secure channel which PID_d^{all} is a list contains aliases of all available D_j .
- **URP-4:** U_i receives $\{PID_u, PID_{gss}, PID_d^{all}, R_2, (C_u^x, R_u^x), Q_i, W_i\}$ and calculates $E_i = MK_u \oplus (PID_u, PID_{gss}, PID_d^{all}, R_2, (C_u^x, R_u^x), Q_i, W_i)$. Finally, U_i securely stores $\{E_i\}$ in the memory.

D. Login and Authentication Phase

The registered U_i and D_j should establish a session key with the help of GSS_n to utilize reliable drone services. We present the overall flow diagram during AKA phase of RLBA-UAV as shown in Fig. 4. Moreover, as shown in Fig. 5, the system participants conduct the following steps for AKA phase.

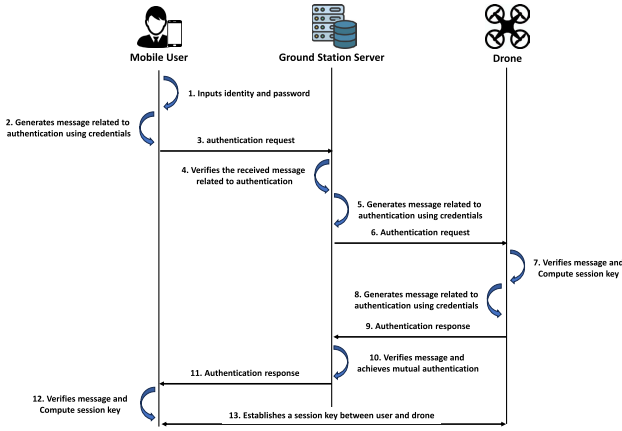


Fig. 4. The overall flow diagram during AKA phase.

LAP-1: U_i first call *UserRegister* function pre-defined in *SC* and obtains E_i in local database. Then, U_i inputs ID_u and PW_u in *MD* and computes $(PID_u, PID_{gss}, PID_d^{all}, R_2, (C_u^x, R_u^x), Q_i, W_i) = E_i \oplus MK_u$. Then, U_i computes $RPW_u = h(PW_u || R_2)$, $X_i = Q_i \oplus h(PID_u || RPW_u || R_u^x)$, and $W_i^* = h(PID_i || RPW_i || R_u^x || X_i)$ and check $W_i^* \stackrel{?}{=} W_i$. If W_i is not correct, the current session is refused, otherwise U_i generates a “random nonce RN_1 , a timestamp TS_1 , and a pair of (C_u^x, R_u^1) from the premise set (C_u^x, R_u^x) ”. After that, U_i computes $M_1 = (RN_1 || PID_u) \oplus h(PID_{gss} || R_u^1 || X_i || TS_1)$ and $Auth_u = h(PID_u || RN_1 || R_u^1 || X_i || TS_1)$ and generates a transaction with the message $\{M_1, Auth_u, C_u^1, TS_1\}$ to trigger *UserAuth* function defined in GSS_n/SC via an open channel.

LAP-2: *SC* selects a pair of (C_u^x, R_u^1) from (C_u^x, R_u^x) and obtains $(RN_1 || PID_u) = M_1 \oplus h(PID_{gss} || R_u^1 || X_i || TS_1)$. Then, *SC* computes $Auth_u^* = h(PID_u || RN_1 || R_u^1 || X_i || TS_1)$ and checks the correctness of message $Auth_u^* \stackrel{?}{=} Auth_u$. If it is not valid, GSS_n terminates the authentication request. Besides, *SC* holds a counter that can tally the failure number of U_i . If the number of U_i approaches to upper bound (e.g. 5 times), U_i will not be able to access the system on the same day. If $Auth_u$ is correct, U_i passes *SC* authentication. GSS_n checks its identity to the corresponding D_j .

LAP-3: GSS_n calls the *DroneAuth* function defined in *SC* with credentials $\{PID_d, PID_u\}$. Then, *SC* selects a random nonce RN_2 , a timestamp TS_2 and a pair of (C_d^x, R_d^1) from the premise set (C_d^x, R_d^x) . *SC* computes $X_i^{new} = h(X_i || RN_1)$, $M_2 = (X_i^{new} || RN_1 || RN_2 || PID_u) \oplus h(PID_d || PID_{gss} || R_d^1 || Z_j || TS_2)$ and $Auth_d = h(PID_u || PID_d || PID_{gss} || RN_2 || R_d^1 || Z_j || TS_2)$. GSS_n sends the message $\{M_2, Auth_d, C_d^1, TS_2\}$ to D_j over an open channel.

LAP-4: After getting the message, D_j checks whether $|TS_2^* - TS_2| \leq \Delta TS_i$. If TS_2 is valid, D_j computes $(PID_d, (C_d^x, R_d^x), Z_j, PID_{gss}) = N_j \oplus MK_d$ and selects a pair of (C_u^1, R_u^1)

from the premise set (C_d^x, R_d^x) . After that, D_j calculates $(X_i^{new} || RN_1 || RN_2 || PID_u) = M_2 \oplus h(PID_d || PID_{gss} || R_d^1 || Z_j || TS_2)$ and $Auth_d^* = h(PID_u || PID_d || PID_{gss} || RN_2 || R_d^1 || Z_j || TS_2)$. Then, D_j verifies the correctness of $Auth_d^* \stackrel{?}{=} Auth_d$. If it is not equal, D_j is terminates the authentication request, otherwise D_j generates a “random nonce RN_3 and a timestamp TS_3 ” and computes $M_3 = (RN_2 || RN_3) \oplus h(PID_u || PID_d || X_i^{new} || TS_3)$, $Auth_{gss} = h(PID_d || PID_{gss} || RN_1 || Z_j || R_d^1 || TS_3)$, $SK_{D-U} = h(RN_1 || RN_2 || RN_3 || PID_u || PID_d)$ and $Auth_{U-D} = h(SK_{D-U} || PID_u || PID_d)$. D_j transmits $\{M_3, Auth_{gss}, Auth_{U-D}, TS_3\}$ to GSS_n .

LAP-5: GSS_n computes $Auth_{gss}^* = h(PID_d || PID_{gss} || RN_1 || Z_j || R_d^1 || TS_3)$ and verifies the correctness of $Auth_{gss}^* \stackrel{?}{=} Auth_{gss}$. If it is not valid, GSS_n terminates the authentication confirmation, otherwise GSS_n authenticates D_j successfully. Finally, GSS_n transmits $\{M_3, Auth_{U-D}, TS_3\}$ to U_i .

LAP-6: After getting the message, U_i computes $X_i^{new} = h(X_i || RN_1)$, $(RN_2 || RN_3) = M_3 \oplus h(PID_u || PID_d || X_i^{new} || TS_3)$, $SK_{U-D} = h(RN_1 || RN_2 || RN_3 || PID_u || PID_d)$, and $Auth_{U-D}^* = h(SK_{U-D} || PID_u || PID_d)$. After that, U_i checks the correctness of $Auth_{U-D}^* \stackrel{?}{=} Auth_{U-D}$. If it is equal, U_i and D_j have successfully established a common session key ($SK_{U-D} = SK_{D-U}$) and are mutually considered to be legitimate entities.

VI. SECURITY ANALYSIS

We introduce the “formal and informal security analyses” to prove the security and robustness of RLBA-UAV.

A. Informal Security Analysis

We further analyze the security and robustness of RLBA-UAV which satisfies the necessary security and functionality properties.

1) *Mutual Authentication:* In RLBA-UAV, all participants perform “secure mutual authentication”. After obtaining the message $\{M_1, Auth_u, C_u^1, TS_1\}$ from U_i , GSS_n verifies the correctness of $Auth_u^* \stackrel{?}{=} Auth_u$. If it is equal, GSS_n authenticates U_i . After receiving the message $\{M_2, Auth_d, C_d^1, TS_2\}$ from GSS_n , D_j checks the correctness of $Auth_d^* \stackrel{?}{=} Auth_d$. If it matches, D_j authenticates GSS_n . After getting the message $\{M_3, Auth_{gss}, Auth_{U-D}, TS_3\}$ from D_j , GSS_n checks $Auth_{gss}^* \stackrel{?}{=} Auth_{gss}$. If it is valid, GSS_n authenticates D_j . After obtaining the message $\{M_3, Auth_{U-D}, TS_3\}$ from GSS_n , U_i verifies the correctness of $Auth_{U-D}^* \stackrel{?}{=} Auth_{U-D}$. If it matches, U_i authenticates GSS_n . Finally, U_i and D_j establish a *SK* for future secure communication. Consequently, RLBA-UAV allows successfully “secure mutual authentication” among U_i , GSS_n , and D_j .

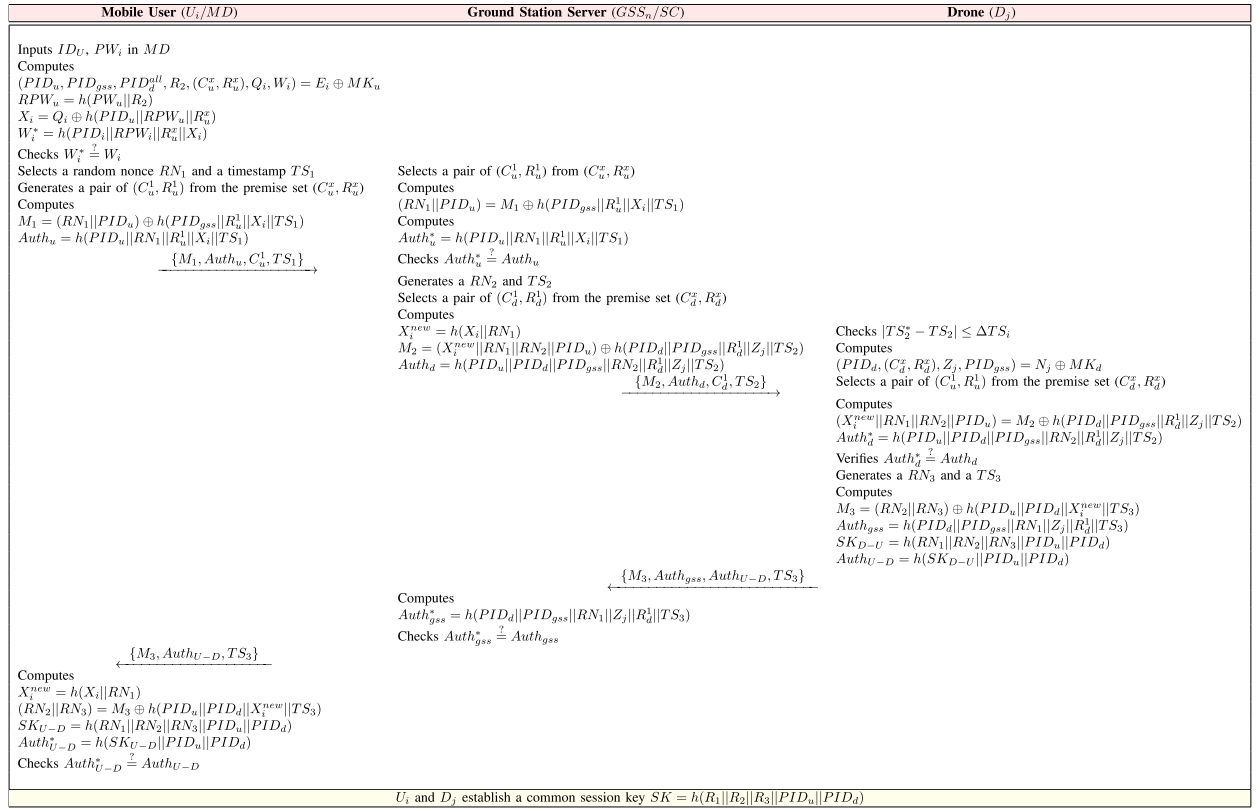


Fig. 5. Summary of login and authentication phase of RLBA-UAV.

2) *MITM Attack*: According to the information given in Section III-A, $\mathcal{A}$ can “resend, eavesdrop, intercept, and delete” the transmitted message during the bidirectional communication among U_i , GSS_n , and D_j in RLBA-UAV. After that, $\mathcal{A}$ tries to get the sensitive data of the legitimate participants. However, $\mathcal{A}$ cannot generate the related messages to authentication because the exchanged messages are protected with “PUF secret values $\{R_u^1, R_d^1\}$ ” and “random nonces $\{RN_1, RN_2, RN_3\}$ ” using XOR and hash functions. Therefore, RLBA-UAV is resilient to MITM attacks since $\mathcal{A}$ cannot obtain the sensitive information related to authentication of the legitimate entities.

3) *Impersonation Attack*: $\mathcal{A}$ attempts to masquerade the legitimate U_i by intercepting the exchanged messages over a public channel. $\mathcal{A}$ should create the authentication request and confirmation messages to impersonate as U_i . Unfortunately, it is deemed difficult to generate the valid messages associated with authentication because $\mathcal{A}$ cannot obtain “PUF secret values $\{R_u^1, R_d^1\}$ ” and temporary secret credentials $\{X_i, Z_j\}$. In addition, the session key is composed of “random nonces $\{RN_1, RN_2, RN_3\}$ ” using hash function. Therefore, RLBA-UAV can prevent this attack because $\mathcal{A}$ cannot successfully generate the authentication messages and establish a SK .

4) *Session Key Disclosure Attack*: Based on the information presented in Section III-A, $\mathcal{A}$ can steal MD and then extract the secret information $\{E_i\}$ stored in the memory using “differential power analysis” [30]. $\mathcal{A}$ should obtain the sensitive data to compute a common session key $SK = h(RN_1 || RN_2 || RN_3 || PID_u || PID_d)$. However, $\mathcal{A}$ is unfeasible to calculate a SK since SK is protected with the “random nonces $\{RN_1, RN_2, RN_3\}$ ” and pseudo identities

“ $\{PID_u, PID_d\}$ ”. Hence, RLBA-UAV is resistant to this attack because $\mathcal{A}$ cannot get the secret credentials related to a session key SK between U_i and D_j .

5) *Physical Capture Attack*: Assume that D_j s are physically captured by $\mathcal{A}$ and extracts the secret credentials $\{N_j\}$ stored in the memory where $N_j = (PID_d, (C_d^x, R_d^x), Z_j, PID_{gss}) \oplus MK_d$. After obtaining these credentials, $\mathcal{A}$ attempts to calculate a common SK between U_i and D_j . However, $\mathcal{A}$ is impossible to compute a $SK = h(RN_1 || RN_2 || RN_3 || PID_u || PID_d)$ without the knowledge of the “PUF secret values $\{R_u^1, R_d^1\}$ ”, “pseudo identities $\{PID_u, PID_d\}$ ”, and “random nonces $\{RN_1, RN_2, RN_3\}$ ”. Moreover, there are distinct, independent and robust for D_j ’s memory because PUF pair $\{(C_d^1, R_d^1)\}$. Therefore, RLBA-UAV demonstrates a significant level of resistance to physical capture attacks given that the output of PUF relies on the inherent physical fluctuations of the IC chip.

B. Formal Security Analysis Using ROR Oracle Model

We evaluate a session key (SK) security of RLBA-UAV using the “ROR oracle model [12]”. We present the necessary queries for this mathematical formal analysis.

In RLBA-UAV, there are three participants: “the mobile user $\Gamma_u^{t_1}$, the ground station server $\Gamma_{gss}^{t_2}$, and the drone $\Gamma_d^{t_3}$, where $\Gamma_u^{t_1}$, $\Gamma_{gss}^{t_2}$, and $\Gamma_d^{t_3}$ are instances t_1^{th} of U , t_2^{th} of GSS , and t_3^{th} of D ”, respectively. Table II presents the necessary queries such as “ $Send(\cdot)$, $Test(\cdot)$, $Execute(\cdot)$, $CorruptMD(\cdot)$, and $Reveal(\cdot)$ ”. We utilize a “PUF function $PUF(\cdot)$ ” and a “hash function $Hash(\cdot)$ ” as a random oracle”.

TABLE II
QUERY AND PURPOSE

Query	Purpose
$Send(\Gamma^t, Msg)$	Under this query, $\mathcal{A}$ can send the messages Msg_i to the Γ^t , and get the response message accordingly.
$CorruptMD(\Gamma_u^{t1})$	This query means as the mobile device stolen attacks where $\mathcal{A}$ can extract the secret information stored in MD .
$Test(\Gamma^t)$	An unbiased coin c is tossed prior to game start. If $\mathcal{A}$ obtains $c = 1$ under the $Test(\cdot)$, it denotes a SK between Γ_u^{t1} and Γ_d^{t3} are fresh. If $\mathcal{A}$ gets the $c = 0$, it means SK is not fresh; otherwise, $\mathcal{A}$ gets a null value ($\perp$).
$Execute(\Gamma_u^{t1}, \Gamma_{gss}^{t2}, \Gamma_d^{t3})$	Under this query, $\mathcal{A}$ attempts the passive/active attacks by eavesdropping the exchanged messages among Γ_u^{t1} , Γ_{gss}^{t2} , and Γ_d^{t3} through an insecure channel.
$Reveal(\Gamma^t)$	Under this query, $\mathcal{A}$ reveals a SK generated between Γ_u^{t1} and Γ_d^{t3} .

Theorem: $Adv_{\mathcal{A}}^{RLBA-UAV}$ denotes the advantages of $\mathcal{A}$ in violating SK security for RLBA-UAV. Therefore, we derive the following:

$$Adv_{\mathcal{A}}^{RLBA-UAV} \leq \frac{q_h^2}{|Hash|} + \frac{q_p^2}{|PUF|} + 2\{C \cdot q_{send}^s \cdot \frac{q_s}{2^{l_p}}\}$$

q_p , q_h , $Hash$, and q_{send} are the “range space of $PUF(\cdot)$ ”, “range space of $h(\cdot)$ ”, “number of $Hash$ query”, and “ $Send(\cdot)$ query”. Besides, l_p , s , and C are the “Zipf’s credentials”.

Proof: We denote the games GM_i ($i \in [0, 4]$). We present that $Adv_{\mathcal{A}, GM_i}^{RLBA-UAV}$ is the probability of $\mathcal{A}$ for winning the GM_i .

Game GM_0 : This game is modeled as “an actual attacks executed by $\mathcal{A}$ ” in RLBA-UAV. The result of GM_0 is as follows:

$$Adv_{\mathcal{A}}^{RLBA-UAV} = |2 \cdot Adv_{\mathcal{A}, GM_0}^{RLBA-UAV} - 1| \quad (1)$$

Game GM_1 : GM_1 denotes that $\mathcal{A}$ carries out an “eavesdropping attack in which the exchanged messages are intercepted among U , GSS , and D using $Execute(\cdot)$ query”. In this game, $\mathcal{A}$ perform the “ $Reveal(\cdot)$ and $Test(\cdot)$ ” queries to reveal SK . The result of the queries decide if $\mathcal{A}$ obtains SK . However, $\mathcal{A}$ does not correctly compute a $SK = h(RN_1 || RN_2 || RN_3 || PID_u || PID_d)$ because he/she cannot obtain the random nonces $\{RN_1, RN_2, RN_3\}$ and pseudo-identities $\{PID_u, PID_d\}$. Therefore, the probability of $\mathcal{A}$ winning GM_1 through eavesdropping remains unchanged. The outcome of this game is outlined below:

$$Adv_{\mathcal{A}, GM_1}^{RLBA-UAV} = Adv_{\mathcal{A}, GM_0}^{RLBA-UAV} \quad (2)$$

Game GM_2 : This game is considered that $\mathcal{A}$ performs the “active/passive attacks” using “ $Hash(\cdot)$ and $Send(\cdot)$ ” queries. $\mathcal{A}$ sends the modified messages to the participants and conducts itself as a legitimate U_i to deduce the results of the “ $Send(\cdot)$ query”. Furthermore, $\mathcal{A}$ endeavors to explore collisions for the hash oracle and tries to imitate messages that each entity is anticipated to authenticate. Since the pseudo-identity and random nonce of the participants are protected by temporary secret credentials $\{X_i, Z_j\}$ and PUF secret values $\{R_u^1, R_d^1\}$ using hash and XOR functions in each

message executing “ $Hash(\cdot)$ and $Send(\cdot)$ queries”, it is not possible to cause a conflict. The probability of discovering collisions in the hash oracle $Hash$ in accordance with the square of the birthday paradox [35] equals $\frac{q_h^2}{2|Hash|}$. The result of GM_2 is as follows:

$$|Adv_{\mathcal{A}, GM_2}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_1}^{RLBA-UAV}| \leq \frac{q_h^2}{2|Hash|} \quad (3)$$

Game GM_3 : This GM_3 is an “extended game to GM_2 which the PUF query $PUF(\cdot)$ in composed of the GM_3 ”. Based on “analogous argument” described in GM_2 , the GM_3 ’s result is below:

$$|Adv_{\mathcal{A}, GM_3}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_2}^{RLBA-UAV}| \leq \frac{q_p^2}{2|PUF|} \quad (4)$$

Game GM_4 : This GM_4 is considered that $\mathcal{A}$ carries out the “differential power analysis” using $CorruptMD(\cdot)$ query. Unfortunately, the secret information $\{E_i\}$ stored in MD memory may be extracted from $\mathcal{A}$ by using “differential power analysis”. However, GM_4 is computationally infeasible for $\mathcal{A}$ to compromise PW_i contained in secret parameters through the $Send(\cdot)$ query without ID_u and R_2 . In addition, given the low probability of $\mathcal{A}$ guessing PUF secret value l_p , it becomes infeasible to differentiate between each properties. The probability of $\mathcal{A}$ succeeding in this game is $\mathcal{A}$ is $\frac{1}{2^{l_p}}$, respectively. Consequently, GM_3 and GM_4 are considered to be indistinguishable in the absence of successful execution of offline password guessing attacks. The result of GM_4 is outlined follows:

$$|Adv_{\mathcal{A}, GM_4}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_3}^{RLBA-UAV}| \leq \{C \cdot q_{send}^s \cdot \frac{q_s}{2^{l_p}}\} \quad (5)$$

After the games are successfully performed, $\mathcal{A}$ attempt to guess the “ c bit for winning the games by performing $Test(\cdot)$ query”. We can obtain the following:

$$Adv_{\mathcal{A}, GM_4}^{RLBA-UAV} = \frac{1}{2} \quad (6)$$

By integrating the “equations (1), (2), and (6)”, we derive the result.

$$\begin{aligned} \frac{1}{2} Adv_{\mathcal{A}}^{RLBA-UAV} &= |Adv_{\mathcal{A}, GM_0}^{RLBA-UAV} - \frac{1}{2}| \\ &= |Adv_{\mathcal{A}, GM_1}^{RLBA-UAV} - \frac{1}{2}| \\ &= |Adv_{\mathcal{A}, GM_1}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_4}^{RLBA-UAV}| \end{aligned} \quad (7)$$

By employing the “triangular inequality” about equations (3), (4), (5), and (7), we can get the subsequent result:

$$\begin{aligned} \frac{1}{2} Adv_{\mathcal{A}}^{RLBA-UAV} &= |Adv_{\mathcal{A}, GM_1}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_4}^{RLBA-UAV}| \\ &\leq |Adv_{\mathcal{A}, GM_1}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_3}^{RLBA-UAV}| \\ &\quad + |Adv_{\mathcal{A}, GM_3}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_4}^{RLBA-UAV}| \\ &\leq |Adv_{\mathcal{A}, GM_1}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_2}^{RLBA-UAV}| \\ &\quad + |Adv_{\mathcal{A}, GM_2}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_3}^{RLBA-UAV}| \\ &\quad + |Adv_{\mathcal{A}, GM_3}^{RLBA-UAV} - Adv_{\mathcal{A}, GM_4}^{RLBA-UAV}| \end{aligned}$$

SUMMARY SAFE	SUMMARY SAFE
DETAILS BOUNDED_NUMBER_OF_SESSIONS TYPED_MODEL PROTOCOL /home/span/span/testsuite/results/sjyu.if	DETAILS BOUNDED_NUMBER_OF_SESSIONS TYPED_MODEL PROTOCOL /home/span/span/testsuite/results/sjyu.if
GOAL As Specified	GOAL as_specified
BACKEND CL-AtSe	BACKEND OFMC
COMMENTS STATISTICS Analysed : 63 states Reachable : 63 states Translation: 0.10 seconds Computation : 0.01 seconds	COMMENTS STATISTICS parseTime: 0.03s searchTime: 2.30s visitedNodes: 1040 nodes depth: 9 plies

Fig. 6. AVISPA results under CL-AtSe and OFMC.

$$\leq \frac{q_h^2}{2|Hash|} + \frac{q_p^2}{2|PUF|} + \{C \cdot q_{send}^s, \frac{q_s}{2^l p}\}. \quad (8)$$

Finally, through the implementation of a scalar multiplication operation on both sides of equation (8) using a multiplier of 2, we get the following: $Adv_{\mathcal{A}}^{RLBA-UAV} \leq \frac{q_h^2}{|Hash|} + \frac{q_p^2}{|PUF|} + 2\{C \cdot q_{send}^s, \frac{q_s}{2^l p}\}$

C. Formal Security Analysis Using AVISPA Simulation

We demonstrate the evidence of the formal security robustness of the cryptographic protocol against various security threats using AVISPA simulation such as “replay and MITM” attacks. We first implement the RLBA-UAV as a programming language “High-Level Protocol Specification Language (HLPSP) [36]”. Then, the simulation commences the analysis of the Intermediate Format (IF) over the two backends: “Constraint Logic-based Attack Searcher (CL-AtSe)” and “On-the-Fly Model Checker (OFMC)”. It is important to highlight that AVISPA is compatible with the DY threat model and encompasses a malicious intruder in the execution of the security protocol with a specific session. We utilize the simulation outcomes achieved through “Security Protocol ANimator (SPAN) [37]”, demonstrating the security attacks and the intruder simulated via a Web-based Graphical User Interface (GUI). The implementation results for CL-AtSe and OFMC are as presented in Fig. 6. Based on the specified security goals, the SAFE output confirms that the proposed RLBA-UAV is secure. Therefore, we have demonstrated that RLBA-UAV is resistant to lethal security attacks.

VII. TESTBED EXPERIMENTS USING MIRACL

We present the testbed experiments to quantify the computation times needed for necessary cryptographic primitives used in our scheme and relevant compared schemes using MIRACL [14] enabled Raspberry PI 4.

We employ two scenarios to measure the computation times of cryptographic primitives. We define T_{sed} , T_{eca} , T_{ecm} , T_{fe} , T_h , and T_{bp} to estimate the execution times needed for “a symmetric key encryption/decryption (e.g. Advanced Encryption Standard (AES)), elliptic curve point addition, elliptic curve scalar point multiplication, fuzzy extractor, hash function (e.g. Secure Hash Algorithm (SHA-256)), and bilinear pairing”. We assumed that $T_{fe} \approx T_{ecm}$.

TABLE III
EXECUTION TIMES (IN MILLISECONDS) FOR A DESKTOP
SETTING BASED ON MIRACL

Operation	Max. time (ms)	Min. time (ms)	Aver. time (ms)
T_{sed}	0.002	0.001	0.001
T_{eca}	0.004	0.002	0.003
T_{ecm}	2.737	0.472	0.522
T_{fe}	2.737	0.472	0.522
T_h	0.149	0.024	0.055

TABLE IV
EXECUTION TIMES (IN MILLISECONDS) FOR A RASPBERRY
PI 4 SETTING BASED ON MIRACL

Operation	Max. time (ms)	Min. time (ms)	Aver. time (ms)
T_{sed}	0.021	0.011	0.012
T_{eca}	0.026	0.017	0.018
T_{ecm}	2.920	2.766	2.848
T_{fe}	2.920	2.766	2.848
T_h	0.643	0.274	0.309

- **Scenario 1:** We have considered a server configuration as outlined: “Model: Desktop, CPU Architecture: 64 bits, Processor: Intel Core i5-10400 @2.90 GHz, Six-core, OS: Ubuntu 18.04.4 LTS with 16 GB memory”. Each cryptographic primitive has been executed 100 times. Subsequently, the minimum, maximum, and average execution times in milliseconds have been measured for each primitive. The experimental results within the server configuration are presented in Table III.
- **Scenario 2:** We have considered Raspberry PI configuration as outlined: “Model: Raspberry PI 4B (2019), CPU Architecture: 64 bit, Processor: 1.5 GHz Quad-core, OS: Ubuntu 20.04.2 LTS with 8 GB memory”. All cryptographic primitives have been executed 100 times. The experimental results based on Raspberry PI 4 configuration are illustrated in Table IV.

VIII. COMPARATIVE ANALYSIS

We present the performance comparative analysis of the RLBA-UAV with related schemes [10], [18], [21], [22], [23] for “computation and communication costs” during AKA phase.

A. Computation Costs

We demonstrate the “computation cost comparison” of RLBA-UAV and related schemes [10], [18], [21], [22], [23]. When the computational cost of the system is reduced, the communication between entities may be more efficiently delivered. We consider the computation costs of the U_i , GSS_n , and D_j to compare with existing schemes. We formalize the types of cryptographic operations essential for the communication channel by analyzing them in milliseconds (ms). We analyze the average execution times for the desktop and Raspberry PI 4 by using the experimental results from the testbed as illustrated in Section IX.

We measured the execution times for GSS_n based on the desktop. In this scenario, we have taken “ $T_{sed} \approx 0.001$, $T_{eca} \approx 0.003$, $T_{ecm} \approx 0.522$, $T_{fe} \approx 0.522$, $T_h \approx 0.055$, and $T_{bp} \approx 3.002$ ” as depicted in Table III. Besides, we measured the

TABLE V
A COMPARATIVE SUMMARY: COMPUTATION COSTS

Scheme	MU_i	GSS_n	D_j
[18]	$16T_h + T_{fe} \approx 7.792$ ms	$8T_h \approx 0.44$ ms	$7T_h \approx 2.163$ ms
[21]	$14T_h + T_{fe} \approx 7.174$ ms	$9T_h \approx 0.495$ ms	$7T_h \approx 2.163$ ms
[22]	$10T_h + T_{fe} \approx 5.938$ ms	$7T_h + 3T_{sed} \approx 0.388$ ms	$7T_h \approx 2.163$ ms
[23]	$6T_h + 2T_{ecm} \approx 7.55$ ms	$8T_h \approx 0.44$ ms	$5T_h + 2T_{ecm} \approx 7.241$ ms
[10]	$5T_h \approx 1.545$ ms	$8T_h \approx 0.44$ ms	$6T_h \approx 1.854$ ms
RLBA-UAV	$9T_h \approx 2.781$ ms	$6T_h \approx 0.33$ ms	$6T_h \approx 1.854$ ms

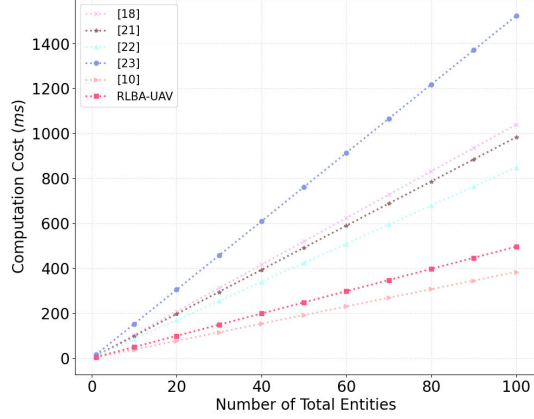


Fig. 7. Computation cost comparison.

TABLE VI
COMPARATIVE ON COMMUNICATION COSTS

Scheme	Total Cost	Number of Messages
[18]	1696 bits	3 messages
[21]	1536 bits	3 messages
[22]	1792 bits	3 messages
[23]	2336 bits	3 messages
[10]	1536 bits	3 messages
RLBA-UAV	1688 bits	4 messages

execution times for U_i and D_j based on Raspberry PI 4. In this scenario, we have taken “ $T_{sed} \approx 0.012$, $T_{eca} \approx 0.018$, $T_{ecm} \approx 2.848$, $T_{fe} \approx 2.848$, $T_h \approx 0.309$, and $T_{bp} \approx 18.294$ ” as depicted in Table IV. Thus, we present the performance result of the computation cost comparison in Fig. 7 Table V. Consequently, RLBA-UAV guarantees a similar computation cost with [10] and has a superior lightweight computation cost compared to related schemes for UAVs [18], [21], [22], [23]. Moreover, RLBA-UAV is resistant to potential security attacks and is superior in terms of fulfilling more security requirements when compared to related schemes.

B. Communication Costs

We demonstrate the “communication cost comparison” of RLBA-UAV and existing schemes [10], [18], [21], [22], [23]. We define that the bit lengths for “the symmetric key encryption, hash function, PUF challenge, identity, timestamp, bilinear pairing, elliptic curve point, and random nonce are 256, 160, 60, 60, 32, 320, 320, and 160 bits”, respectively.

In AKA phase of RLBA-UAV, the exchanges messages $\{M_1, Auth_u, C_u^1, TS_1\}$, $\{M_2, Auth_d, C_d^1, TS_2\}$, $\{M_3, Auth_{gss}, Auth_{U-D}, TS_3\}$, and $\{M_3, Auth_{U-D}, TS_3\}$ require $(160 + 160 + 60 + 32 = 412$ bits), $(160 + 160 + 60 + 32 = 412$ bits), $(160 + 160 + 160 + 32 = 512$ bits), and $(160 + 160 + 32 = 352$ bits). Thus, RLBA-UAV has an total communication cost of 1688 bits. In Table VI and Fig. 8, although RLBA-UAV has a slightly higher cost compared to

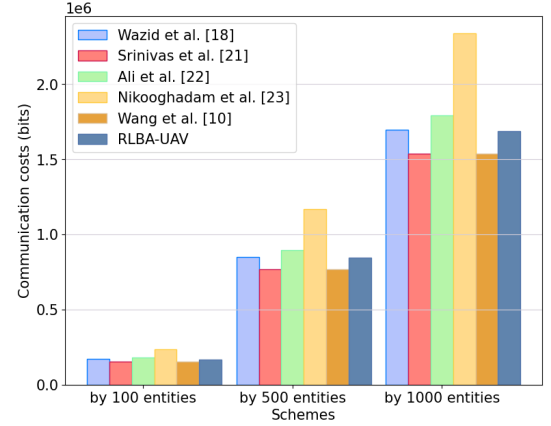


Fig. 8. Communication cost comparison.

TABLE VII
SIMULATION PARAMETERS

Parameters	Values
Simulation tool	NS-3 (3.29)
Operating system	Ubuntu 20.04.6 LTS
Routing protocol	AODV
Simulation time	1800s
Network	IEEE 802.11p
Area	400m×400m×100m
Mobility of D_j	Random (0-45 m/s)
Number of U_i	5 (for scenarios 1, 2, 3)
	10 (for scenarios 1, 2, 3)
	15 (for scenarios 1, 2, 3)
Number of D_j	20 (for scenario 1)
	30 (for scenario 2)
	40 (for scenario 3)

schemes [10], [21], it has proved that RLBA-UAV provides lightweight communication costs due to its lower costs compared to schemes [18], [22], [23].

IX. NS-3 IMPLEMENTATION

To evaluate the effectiveness and accessibility of the proposed approach, we employed NS-3 [13] on an Ubuntu 20.04.6 LTS platform using an “Intel Core i5-10400 CPU clocked at 2.90 GHz”. In our simulation setup, GSS_n remains stationary at the central position, while D_j is positioned randomly within a “distance spanning from 10 to 350m from GSS_n ”. Additionally, we configured the mobility of U_i to reach a “maximum speed of 3 m/s within a radius of 0 to 300m around GSS_n ”. Further details regarding the experimental settings can be found in Table VII.

Within our proposed framework, the elements U_i , D_j , and GSS_n participate in the transmission of authentication messages. These messages encompass $Msg_1 = \{M_1, Auth_u, C_u^1, TS_1\}$, spanning 51.5 bytes, $Msg_2 = \{M_2, Auth_d, C_d^1, TS_2\}$ with a size of 51.5 bytes, $Msg_3 = \{M_3, Auth_{gss}, Auth_{U-D}, TS_3\}$ totaling 64 bytes, and $Msg_4 = \{M_3, Auth_{U-D}, TS_3\}$ comprising 32 bytes. The latency and data transfer rate of these messages are evaluated across three distinct test scenarios.

A. End-to-End Delay

We assess the overall latency from source to destination known as end-to-end latency, which indicates the average time taken for data packets to traverse.

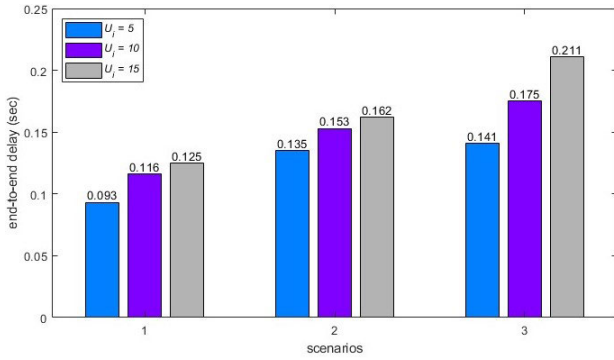


Fig. 9. End-to-end delay.

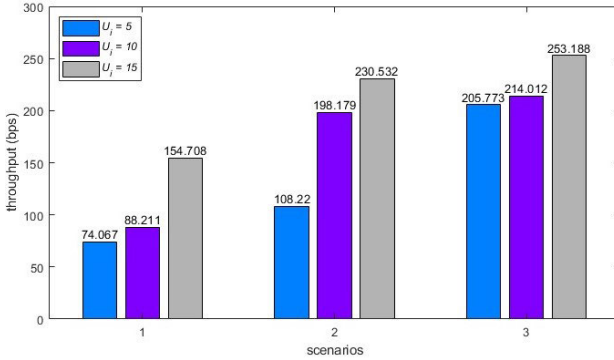


Fig. 10. Throughput.

This calculation involves computing the summation of the differences between reception and transmission times for each packet, divided by the total packet count (P_t). Mathematically, it is expressed as $\sum_{y=0}^{P_t} (T_{R_y} - T_{S_y}) / P_t$, where " P_t " represents the total packet count, T_{R_y} denotes the reception time of the i th packet, and T_{S_y} signifies the transmission time of the i th packet". Fig. 9 demonstrates the observed end-to-end delay across three distinct simulated scenarios. With an increasing number of entities, packet forwarding escalates, potentially leading to heightened end-to-end delays due to resource contention and traffic congestion. Consequently, as the quantity of U_i and D_j entities rises, the end-to-end delay expands, corresponding to the decreased spatial separation between entities.

B. Throughput

Throughput serves as a measure quantifying the rate of transmitted data bits over a given period within a communication network. The computation of throughput is represented by the formula $(R \times |\eta|) / T_R$, where R , $|\eta|$, and T_R represent "the number of packets received, the size of each packet, and the total duration in seconds", respectively. The simulation results presented in Fig. 10 display the observed throughput in the proposed framework over a simulation period of 1800 seconds. The analysis illustrates a positive correlation between network throughput and the quantity of exchanged messages, indicating an escalation in throughput as message volume increases.

X. CONCLUSION

We prove that the related schemes are vulnerable to impersonation and session key disclosure attacks and lack secure

mutual authentication. Therefore, we propose a robust and lightweight blockchain-based AKA scheme for PUF-enabled UAVs (RLBA-UAV) to improve the security flaws of existing schemes. We demonstrated the session key security of RLBA-UAV by performing the ROR oracle model. In addition, we proved that RLBA-UAV is secure to replay and MITM attacks via AVISPA implementation. We demonstrated the impact on diverse network performance parameters through an RLBA-UAV implementation using NS-3 simulation in compliance with the IEEE 802.11p standard network. Finally, we evaluated the performance comparison analysis of RLBA-UAV and the related schemes for UAVs in terms of communication and computation costs. Consequently, we proved that RLBA-UAV provides superior security than related schemes for UAVs and also guarantees better lightweight communication and computation costs than existing schemes.

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¹Trademarked.